WORKED SOLUTION

STEP II 2017, Question 13

Mechanics • Game show

STEP II 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q13

The key idea

In every version the door opens at attempt k exactly when the first $k-1$ attempts fail and the kth succeeds, so the probability is a product of "fail" factors times one "success" factor. What changes between versions is how many keys are in the bag at each draw. Once the probabilities are written down, the expectations are sums: a differentiated geometric series in (i), an arithmetic series in (ii), and in (iii) a telescoping sum that leaves behind a piece of the harmonic series.

Solution

Throughout, let X be the number of the attempt at which the door opens, and let $q = 1 - \frac{1}{n}$.

Part (i)

Each draw is from the full bag of n keys, independently of earlier draws because the key is replaced. So each attempt succeeds with probability $\frac{1}{n}$, and

$$\mathrm{P}(X = k) = q^{k-1} \cdot \frac{1}{n}, \qquad k = 1, 2, 3, \ldots$$

By the binomial expansion, valid since $0 < q < 1$,

$$(1-q)^{-2} = 1 + 2q + 3q^2 + \cdots = \sum_{k=1}^{\infty} k q^{k-1}.$$

Hence

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} k\, q^{k-1} \frac{1}{n} = \frac{1}{n}(1-q)^{-2} = \frac{1}{n} \cdot n^2 = n.$$

The expected number of attempts is n.

Part (ii)

Now the bag loses one wrong key after each failure, so at attempt j there are $n-j+1$ keys. For the door to open at attempt k we need $k-1$ failures and then a success:

$$\mathrm{P}(X=k)=\frac{n-1}{n}\cdot\frac{n-2}{n-1}\cdots\frac{n-k+1}{n-k+2}\cdot\frac{1}{n-k+1}=\frac{1}{n},$$

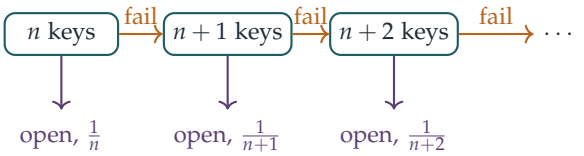
since the product telescopes. (This is also clear directly: drawing keys one by one without replacement puts the correct key in a uniformly random position among n.) So X is uniform on $\{1,2,\ldots,n\}$ and

$$\mathrm{E}(X)=\frac{1}{n}\sum_{k=1}^{n}k=\frac{1}{n}\cdot\frac{n(n+1)}{2}=\frac{n+1}{2}.$$

Part (iii)

The probability. The bag starts with n keys and gains one wrong key after each failure, and always contains exactly one correct key. So at attempt j there are $n+j-1$ keys, and

$$\mathrm{P}(\text{attempt } j \text{ fails} \mid \text{earlier attempts failed})=\frac{n+j-2}{n+j-1}.$$



Multiplying the conditional probabilities for $k-1$ failures and then a success,

$$\begin{aligned}\mathrm{P}(X=k)&=\frac{n-1}{n}\cdot\frac{n}{n+1}\cdots\frac{n+k-3}{n+k-2}\cdot\frac{1}{n+k-1}\\&=\frac{n-1}{n+k-2}\cdot\frac{1}{n+k-1}=\frac{n-1}{(n+k-1)(n+k-2)},\end{aligned}$$

because each numerator cancels the previous denominator, leaving the first numerator $n-1$ and the last fail-denominator $n+k-2$. For $k=1$ there are no fail factors, and the formula gives $\frac{n-1}{n(n-1)}=\frac{1}{n}$, as it should. This is the required result.

The door does open. Before computing the mean it is worth checking that X is finite with probability 1. Since $\dfrac{n-1}{(n+k-1)(n+k-2)}=(n-1)\left(\dfrac{1}{n+k-2}-\dfrac{1}{n+k-1}\right)$, the sum telescopes:

$$\sum_{k=1}^{N}\mathrm{P}(X=k)=(n-1)\left(\frac{1}{n-1}-\frac{1}{n+N-1}\right)\to 1\quad\text{as } N\to\infty.$$

The expectation. We need $\sum k\,\mathrm{P}(X=k)$, so we split $\dfrac{k}{(n+k-1)(n+k-2)}$ into partial fractions. Writing

$$k = A(n+k-2) + B(n+k-1)$$

and putting $k = 1-n$ gives $1-n = -A$, so $A = n-1$; putting $k = 2-n$ gives $B = 2-n$. Hence

$$k\,\mathrm{P}(X=k) = (n-1)\left(\frac{n-1}{n+k-1} - \frac{n-2}{n+k-2}\right).$$

Ask yourself This doesn't telescope neatly, because the coefficients $n-1$ and $n-2$ differ. So what is left over?

Let S_N be the sum of the first N terms. In the first fraction the denominators run over $m = n, \ldots, n+N-1$; in the second over $m = n-1, \ldots, n+N-2$. On the common range $m = n, \ldots, n+N-2$ each $\frac{1}{m}$ appears with coefficient $(n-1)-(n-2) = 1$, and the two ends are left over:

$$S_N = (n-1)\left(\sum_{m=n}^{n+N-2}\frac{1}{m} + \frac{n-1}{n+N-1} - \frac{n-2}{n-1}\right).$$

The last two terms are bounded (the middle one is positive and the last is a fixed number), while

$$\sum_{m=n}^{n+N-2}\frac{1}{m} = \sum_{m=1}^{n+N-2}\frac{1}{m} - \sum_{m=1}^{n-1}\frac{1}{m} \to \infty$$

as $N \to \infty$, since the harmonic series diverges and we subtract only a fixed finite amount. As $n-1 > 0$, $S_N \to \infty$. All the terms $k\,\mathrm{P}(X=k)$ are positive, so the series defining $\mathrm{E}(X)$ diverges to infinity: the expected number of attempts is infinite, even though the door is certain to open eventually.

Checks

- For $n = 2$ the three answers are 2, $\frac{3}{2}$ and ∞. Version 2 is the most efficient, as it must be, since it never retries a key; version 1 is in between.
- In (iii) with $n = 3$ the formula gives $\frac{1}{3}, \frac{1}{6}, \frac{1}{10}$ for $k = 1, 2, 3$; directly, $\frac{1}{3}$, $\frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}$ and $\frac{2}{3}\cdot\frac{3}{4}\cdot\frac{1}{5} = \frac{1}{10}$. A computer simulation of 2×10^5 games gave frequencies 0.334, 0.168 and 0.099.
- Also for $n = 3$, the partial sum S_N with $N = 10^4$ is about 15.6, and it keeps growing like $2\ln N$: there is no limit.

Common slips

- In (iii), taking the bag size at attempt k to be $n+k$ rather than $n+k-1$: only $k-1$ keys have been added before the kth draw.

- Concluding that the door might never open because the mean is infinite. The probabilities sum to 1; it is only the tail, decaying like $1/k^2$, that is too heavy for $\sum k\,\mathrm{P}(X = k)$ to converge.
- Telescoping the partial fractions as if both coefficients were equal and getting a finite answer.

Beyond STEP

Version 3 gives a random variable that is finite with probability 1 but has infinite mean, because $\mathrm{P}(X = k)$ behaves like $1/k^2$. Heavy-tailed distributions like this, and the famous St Petersburg paradox, show why first-year probability courses insist that $\mathrm{E}(X)$ exists only when $\sum k\,\mathrm{P}(X = k)$ converges. The bag that gains a wrong key after each failure is a relative of Pólya's urn, a standard model of "rich get richer" processes. Grimmett and Welsh, *Probability: An Introduction*, covers expectation of discrete random variables and these subtleties.

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